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Course: CPEN 208-Software Engineering

Project 1

This project implements a web application, a student portal for the 'School of Software Engineering', built with using Next.js 14 which allows users to register, log-in on a dashboard created with a PostgreSQL relational database of the following:

1. Student Personal information
2. Student Fees Payments
3. Course Enrollment
4. Lectures to Course Assignment
5. Lectures to TA assignment.

1. User System

- Registration: Users can create accounts with name, email, and password
- Login: Secure login with forgot password option
- Dashboard: Shows user's personal information and courses

2. Dashboard Features

- Welcome Message: Shows "Welcome, [User Name]"
- Courses: Displays enrolled courses with grades and lecturers
 - CPEN201: Database (Dr. John Assiamah)
 - CPEN208: Software Engineering (Dr. John Assiamah)
 - CPEN204: Algorithms (Dr. Sarah Mensah)
 - CPEN302: Database Systems (Dr. Michael Addo)
- Fees: Shows payment status and remaining balance
- Advisor: Dr. John Assiamah from Computer Engineering

3. Design

- University Logo: Added UG logo to login and registration pages
- Clean Interface: Simple, easy to use design

Technical Details

- Database: PostgreSQL with user information
- Authentication: Users stay logged in until they logout
- API: Backend handles user data and dashboard information

Files Created/Modified

- Registration page with password confirmation
- Login page with forgot password
- Dashboard with dynamic user data
- API endpoints for user management
- Database integration

Technologies Used:

- Next.js 14 (App Router)
- PostgreSQL (pgAdmin used for management)
- Prisma ORM
- Tailwind CSS
- Git and GitHub

Database Tables:

- student
- courses
- lecturer
- TA
- course_enrollment
- student_fees_payments
- user

A working student portal where users can register, login, and view their academic information in a clean, professional interface.

GitHub Repository

https://github.com/SteelLee123/22225243_Lab

Backup File:

The file db_backup.sql is included in the database/ folder.

Conclusion

This project demonstrates how to build and connect a modern web app to a relational database system, simulating a basic academic portal with real user management.